

Chapter 5 - Sets: Basic Definitions

1 pts

1. Determine the power set of $\{\emptyset, 1, \{1\}\}$.

5 pts

2. Determine the cardinality of the following sets:

(a) The empty set.

(b) The power set of the empty set.

(c) The power set of the power set of the empty set.

(d) The power set of $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$.

(e) $\{1, \{1\}, \{1, 2\}\}$

8 pts

3. Mark each statement as ☐ True or ☐ False by completely shading in the box corresponding to your answer.

- (a) ☐ True ☐ False For any set A , $A \subseteq \mathcal{P}(A)$.
- (b) ☐ True ☐ False For any set A , $A \in \mathcal{P}(A)$.
- (c) ☐ True ☐ False For any finite set A , $|\mathcal{P}(A)| > |A|$.
- (d) ☐ True ☐ False If $n \in \mathbb{N}$, then $n + 1 \in \mathbb{N}$.
- (e) ☐ True ☐ False If $n \in \mathbb{N}$, then $n - 1 \in \mathbb{N}$.
- (f) ☐ True ☐ False If $n \in \mathbb{Z}$, then $n + 1 \in \mathbb{Z}$.
- (g) ☐ True ☐ False If $n \in \mathbb{Z}$, then $n - 1 \in \mathbb{Z}$.
- (h) ☐ True ☐ False $\mathbb{Z} \subseteq \mathbb{Q}$.

Chapter 6 - Set Operations

4 pts

1. Let $A = \{1, 2, 3, 5, 6, 7, 9\}$ and $B = \{1, 3, 4, 6, 8, 9\}$. Find

(a) $A \cap B$

(b) $A \cup B$

(c) $A - B$

(d) $B - A$

2 pts

2. Use the rules of set algebra to verify $A \oplus B = (A \cup B) - (A \cap B)$.

2 pts

3. Let $A = \{1, 2, 3\} \times \{1, 2, 3, 4\}$. List the elements of the set $B = \{(s, t) \in A \mid s < t\}$.

3 pts

4. Let $A = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\} \times \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$.

(a) What is the cardinality of A ?

(b) What is the cardinality of $A \times A$?

(c) What is the cardinality of $B = \{s \mid (s, s^2) \in A\}$?

4 pts

5. Mark each statement as

True

 or

False

 by completely shading in the box corresponding to your answer.

(a)

True

False

 For all sets A, B, C , $A \times (B \cap C) = (A \times B) \cap (A \times C)$.

(b)

True

False

 For all sets A, B, C , $A \times (B \cup C) = (A \times B) \cup (A \times C)$.

(c)

True

False

 For all sets A, B, C, D , $(A \cap B) \times (C \cap D) = (A \times C) \cap (B \times D)$.

(d)

True

False

 For all sets A, B, C, D , $(A \cup B) \times (C \cup D) = (A \times C) \cup (B \times D)$.

Chapter 7 - Styles of Proof

3 pts

1. Give a direct proof that the sum of an even integer and an odd integer is odd.

Hint: Start by letting m be an even integer and letting n be an odd integer. That means $m = 2k$ for some integer k and $n = 2j + 1$ for some integer j . You are interested in $m + n$, so add them up and see what you get. Why is the thing you get an odd integer (think about the definition of *odd*)?

3 pts

2. Give a direct proof that the sum of two odd integers is even.

3 pts

3. Give a proof by contradiction that if $3n + 2$ is odd, then n is odd.

Hint: The first line of the proof should be something like "Suppose $3n + 2$ is odd **and** n is even."

2 pts

4. Give an example of a predicate $P(n)$ about positive integers n , such that $P(n)$ is true for every positive integer from 1 to one billion, but which is never-the-less not true for all positive integers.

Hint: there is a really simple choice possible for the predicate $P(n)$.